

D845

General Physics I

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

Two warehouse workers push a 50 kg crate across a nearly frictionless floor. One pushes east with 300 N; the other, working from the opposite side, pushes west with 100 N. The crate is painted bright orange and was shipped from a depot 40 km away. What is the crate's acceleration?

- A. It cannot be found without the distance the crate travels.
- B. 4 m/s² east
- C. 8 m/s² east
- D. 6 m/s² east

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

On a rainy afternoon, a 20,000 kg delivery truck collides head-on with a 1,000 kg compact car. The truck driver has 25 years of experience. During the impact, how does the force the car exerts on the truck compare with the force the truck exerts on the car?

- A. The car exerts no force, because the truck keeps moving forward.
- B. Equal in size and opposite in direction.
- C. The car's force is smaller because the wet road reduced it.
- D. The truck's force on the car is 20 times larger.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

In a collision between a 20,000 kg truck and a 1,000 kg car, which quantity is different for the two vehicles?

- A. The time each vehicle is in contact with the other.
- B. The size of the impulse each receives.
- C. The size of their accelerations; the car's is 20 times larger.
- D. The size of the forces they exert on each other.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

At a science fair, a teacher whirls a 0.5 kg ball on a 2 m string in a horizontal circle at 4 m/s. The string is rated to 100 N. Ignoring gravity, what tension does the string supply?

- A. 4 N
- B. 1 N
- C. 100 N
- D. 16 N

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

A ball is whirled in a horizontal circle on a string. The string suddenly snaps. Seen from above, which way does the ball move?

- A. In a straight line tangent to the circle.
- B. Inward, toward the center.
- C. Straight outward, away from the center.
- D. It keeps curving along the circle for a while.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

An office manager slides a 40 kg box labeled fragile across a tile floor at a steady 1.5 m/s. The coefficient of kinetic friction is 0.25. Using $g = 9.8 \text{ m/s}^2$, what is the friction force on the box?

- A. 98 N
- B. 392 N
- C. 147 N
- D. 10 N

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

At a four-way stop, a 2,000 kg SUV moving at 10 m/s rear-ends a stopped 1,000 kg car. The SUV's brakes had just been serviced. The vehicles lock together. What is their speed just after impact?

- A. 5 m/s
- B. 10 m/s
- C. About 6.7 m/s
- D. About 3.3 m/s

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

A 1,500 kg car hits a stationary 500 kg car. They stick together and move off at 6 m/s. How fast was the first car going before impact?

- A.** 4.5 m/s
- B.** 6 m/s
- C.** 24 m/s
- D.** 8 m/s

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A hiker knocks a 2 kg rock off a 20 m cliff. The rock is granite and about the size of a fist. Ignoring air resistance and using $g = 9.8 \text{ m/s}^2$, how fast is it moving just before it hits the ground?

- A.** 28.0 m/s
- B.** 392 m/s
- C.** About 14.0 m/s
- D.** About 19.8 m/s

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A radio station with a 50,000-watt transmitter broadcasts at 100 MHz ($1.00 \times 10^8 \text{ Hz}$) and reaches listeners 80 km away. Radio waves travel at $3.00 \times 10^8 \text{ m/s}$. What is the wavelength of the signal?

- A.** 0.33 m
- B.** 80 km
- C.** 3.0 m
- D.** $3.0 \times 10^{16} \text{ m}$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

A sound wave passes from air into water, where it travels about four times faster. Which property of the wave stays the same?

- A.** All of its properties, since waves never change between media.
- B.** Its wavelength.
- C.** Its speed.
- D.** Its frequency.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A research probe built by a university team at a cost of \$4 million weighs 1,200 N on Earth's surface. It is placed in orbit where its distance from Earth's center is three times Earth's radius. What gravitational force acts on it there?

- A.** 10,800 N
- B.** About 133 N
- C.** 400 N
- D.** 0 N, because objects in orbit are weightless.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

Two warehouse workers push a 50 kg crate across a nearly frictionless floor. One pushes east with 300 N; the other, working from the opposite side, pushes west with 100 N. ~~The crate is painted bright orange and was shipped from a depot 40 km away.~~ What is the crate's acceleration?

Target word: acceleration from net force **Decides:** Net force = 300 N – 100 N = 200 N east.

A	Camouflage	The 40 km shipping distance has nothing to do with $F = ma$.
B	Correct	200 N / 50 kg = 4 m/s ² in the direction of the net force.
C	Inverted operation	Adds opposing forces (400 N) instead of subtracting.
D	Dropped element	Uses only the 300 N push and ignores the opposing worker.

2. Answer B Concept 2

~~On a rainy afternoon,~~ a 20,000 kg delivery truck collides head-on with a 1,000 kg compact car. ~~The truck driver has 25 years of experience.~~ During the impact, how does the force the car exerts on the truck compare with the force the truck exerts on the car?

Target word: compare the two contact forces **Decides:** The forces form a third-law pair.

A	Absolute language	Every push comes with an equal push back, whatever the motion afterward.
B	Correct	Newton's third law: interaction forces are always equal and opposite.
C	Camouflage	Road conditions do not break the third-law pair.
D	Confusion set	Mixes up force with acceleration; the car's acceleration is 20 times larger, not the force.

3. Answer C Concept 2 · transfer

In a collision between a 20,000 kg truck and a 1,000 kg car, which quantity is different for the two vehicles?

Target word: what differs — reverse of the equal-force question **Decides:** Equal forces on unequal masses give unequal accelerations.

A	Adjacent category	They touch for the same moment; contact time is shared.
B	Confusion set	Impulse = force × time, equal for both; only the velocity change differs.
C	Correct	$a = F/m$ with the same F ; the car has 1/20 the mass.
D	Inverted operation	That is exactly what the third law says is equal.

4. Answer A Concept 3

~~At a science fair,~~ a teacher whirls a 0.5 kg ball on a 2 m string in a horizontal circle at 4 m/s. ~~The string is rated to 100 N.~~ Ignoring gravity, what tension does the string supply?

Target word: centripetal force **Decides:** $F = mv^2/r$.

A	Correct	$0.5 \times 4^2 / 2 = 0.5 \times 16 / 2 = 4$ N.
B	Dropped element	$0.5 \times 4 / 2$ forgets to square the speed.
C	Camouflage	The string's rating is its limit, not the tension in use.
D	Inverted operation	Multiplies by the radius instead of dividing.

5. Answer A Concept 3 • transfer

A ball is whirled in a horizontal circle on a string. The string suddenly snaps. Seen from above, which way does the ball move?

Target word: motion once the inward force is gone **Decides:** No net force means straight-line motion along the velocity at that instant.

A	Correct	Inertia keeps the ball moving in the direction it was already going.
B	Inverted operation	The inward pull was the string; once cut, nothing pulls inward.
C	Confusion set	Assumes a real outward “centrifugal” force; there is none in an inertial frame.
D	Dropped element	Curving requires an inward force, which disappeared with the string.

6. Answer A Concept 4

An office manager slides a 40 kg box ~~labeled fragile~~ across a tile floor ~~at a steady 1.5 m/s~~. The coefficient of kinetic friction is 0.25. Using $g = 9.8 \text{ m/s}^2$, what is the friction force on the box?

Target word: kinetic friction force **Decides:** $f = \mu_k \times N$, with $N = mg = 392 \text{ N}$.

A	Correct	$0.25 \times 40 \times 9.8 = 98 \text{ N}$.
B	Adjacent category	That is the normal force, not the friction force.
C	Camouflage	Multiplies by the speed; kinetic friction does not depend on speed.
D	Dropped element	0.25×40 uses mass, not normal force; g is missing.

7. Answer C Concept 5

~~At a four-way stop,~~ a 2,000 kg SUV moving at 10 m/s rear-ends a stopped 1,000 kg car. ~~The SUV’s brakes had just been serviced.~~ The vehicles lock together. What is their speed just after impact?

Target word: shared final velocity **Decides:** Momentum $2,000 \times 10 = 20,000 \text{ kg}\cdot\text{m/s}$ is conserved over 3,000 kg.

A	Confusion set	Averages the two speeds instead of conserving momentum.
B	Dropped element	Ignores the added mass of the car.
C	Correct	$20,000 / 3,000 \approx 6.67 \text{ m/s}$.
D	Wrong stem number	Uses the car’s mass ($1,000 \times 10$) as the starting momentum.

8. Answer D Concept 5 • transfer

A 1,500 kg car hits a stationary 500 kg car. They stick together and move off at 6 m/s. How fast was the first car going before impact?

Target word: solve for initial speed **Decides:** Final momentum $= 2,000 \times 6 = 12,000 \text{ kg}\cdot\text{m/s}$.

A	Inverted operation	Flips the mass ratio ($6 \times 1,500 / 2,000$), making the car slower before than after.
B	Wrong stem number	The final speed given in the stem.
C	Adjacent category	Divides by the struck car’s mass instead of the moving car’s.
D	Correct	$12,000 / 1,500 = 8 \text{ m/s}$.

9. Answer D Concept 6

A hiker ~~knocks~~ a 2 kg rock off a 20 m cliff. ~~The rock is granite and about the size of a fist.~~ Ignoring air resistance and using $g = 9.8 \text{ m/s}^2$, how fast is it moving just before it hits the ground?

Target word: speed at the bottom **Decides:** $v = \sqrt{2gh}$; mass cancels.

A	Camouflage	Leaves the 2 kg mass in the formula; mass cancels from both sides.
B	Dropped element	Stops at v^2 and skips the square root.
C	Inverted operation	Uses $v^2 = gh$, dropping the factor of 2 that comes from $\frac{1}{2}mv^2$.
D	Correct	$\sqrt{(2 \times 9.8 \times 20)} = \sqrt{392} \approx 19.8 \text{ m/s}$.

10. Answer C Concept 7

A radio station ~~with a 50,000 watt transmitter~~ broadcasts at 100 MHz ($1.00 \times 10^8 \text{ Hz}$) ~~and reaches listeners 80 km away.~~ Radio waves travel at $3.00 \times 10^8 \text{ m/s}$. What is the wavelength of the signal?

Target word: wavelength **Decides:** $\lambda = v / f$.

A	Inverted operation	Divides frequency by speed.
B	Camouflage	The broadcast range, not the wavelength.
C	Correct	$3.00 \times 10^8 / 1.00 \times 10^8 = 3.0 \text{ m}$.
D	Confusion set	Multiplies $v \times f$, mixing up the formula for speed with the one for wavelength.

11. Answer D Concept 7 • transfer

A sound wave passes from air into water, where it travels about four times faster. Which property of the wave stays the same?

Target word: the invariant across a boundary **Decides:** The source sets frequency.

A	Absolute language	Speed and wavelength both change at the boundary.
B	Confusion set	Wavelength grows about four times because $v = f\lambda$.
C	Inverted operation	The stem says speed changes.
D	Correct	Frequency is fixed by the source; speed changes, so wavelength changes.

12. Answer B Concept 8

A research probe ~~built by a university team at a cost of \$4 million~~ weighs 1,200 N on Earth's surface. It is placed in orbit where its distance from Earth's center is three times Earth's radius. What gravitational force acts on it there?

Target word: gravity at $3\times$ the radius **Decides:** Force scales as $1/r^2$: $1/3^2 = 1/9$.

A	Inverted operation	Multiplies by 9; gravity weakens with distance.
B	Correct	$1,200 / 9 \approx 133 \text{ N}$.
C	Dropped element	Divides by 3, dropping the square.
D	Absolute language	Orbiting objects are in free fall, but gravity still acts on them.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	7	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	3	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	9	
Absolute language	Always, never, guarantee — where judgment applies.	3	
Confusion set	Every option is real; the test is exact role-matching.	6	
Camouflage	Answers a stem fact that was never the question.	6	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Newton's second law uses net force	Acceleration = net force / mass, where net force is the vector sum of all forces.	Plugging in one applied force, or adding forces that point in opposite directions.	Two or more forces acting on one object in the same stem.
2	Newton's third law pairs	Action and reaction forces are equal in size, opposite in direction, and act on different objects.	Believing the bigger or faster object pushes harder.	A collision or push between two objects of very different mass.
3	Centripetal force	Circular motion needs a net inward force $F = mv^2/r$; "centrifugal force" is not a real outward push in an inertial frame.	Thinking an outward force flings objects away when the inward force is removed.	A string, curve, or orbit plus a speed and a radius.
4	Kinetic friction	Kinetic friction = $\mu_k \times$ normal force; it does not depend on speed.	Using mass instead of normal force, or scaling friction with speed.	A coefficient μ and a sliding object.
5	Momentum conservation in collisions	Total momentum before a collision equals total momentum after; objects that stick share one final velocity.	Averaging speeds or conserving kinetic energy in an inelastic collision.	Objects that "lock together" or "stick."
6	Conservation of mechanical energy	With no friction, $mgh = \frac{1}{2}mv^2$, so $v = \sqrt{2gh}$ and mass cancels.	Thinking heavier objects fall faster or forgetting the square root.	A drop height and a question about speed at the bottom.
7	Wave speed, frequency, wavelength	$v = f\lambda$; the source sets frequency, the medium sets speed, so wavelength adjusts.	Thinking frequency changes when a wave enters a new medium.	A wave crossing a boundary, or two of v , f , λ given.
8	Inverse-square gravity	Gravitational force varies as $1/r^2$, with r measured from the planet's center.	Using $1/r$, or thinking orbiting objects feel no gravity.	A distance given as a multiple of the planet's radius.

Course scope (public description): This General Physics course covers fundamental concepts, including Newton's Laws, forces, motion, energy, waves, electricity, and magnetism, with real-world applications and insights into relativity and quantum theory. Learners will study measurement, forces and motion, Newton's Laws, centrifugal and centripetal forces, friction, gravity, momentum, collisions, vectors, wave motion, energy, thermodynamics, and electromagnetic waves. Skills developed include scientific literacy, physical science application, systems thinking, and scientific reasoning.